Submitted in partial fulfillment of requirement for award the degree of

**BACHELOR OF COMPUTER APPLICATIONS**

To the

# A Project Report

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**Serkkadu, Vellore-632115**

# Cricket Player Performance Tracking

# Application : CRICKSTATS

## User Story

As a Salesforce Administrator, my goal is to develop a complete player performance tracking system within Salesforce that records and evaluates detailed match-by-match data of cricket players. The application will use custom objects such as Player and Match Performance with properly structured fields and relationships to maintain accurate and well-organized data.

By utilizing master-detail and lookup relationships, the system will ensure efficient data access and reliability. This solution will help monitor both individual and team performances, generate meaningful reports and analytics, and support the effective management of player statistics, ultimately improving strategic decisions and player development.

# 1. INTRODUCTION

## 1.1 Project Overview

CRICKSTATS is a Salesforce-based system created to monitor, organize, and evaluate the performance of cricket players for every match they play.

The application makes use of custom objects, defined relationships, dashboards, and automation features to deliver well-structured insights into player performance.

This platform allows coaches, analysts, and administrators to document player statistics, create analytical reports, and assist in making informed decisions for team strategies.

## 1.2 Purpose

The primary objectives of this project are:

- To convert player performance records into a digital format
- To enhance the reliability and availability of cricket statistics
- To deliver real-time data analysis and insights
- To assist in player improvement and team strategy planning
- To streamline and automate data management through Salesforce
- If you want, I can also rewrite the Ideation Phase (Problem Statement table) in different wording so your whole document becomes plagiarism-free for submission.

# 2. IDEATION PHASE

## 2.1 Problem Statement (Customer Problem Statement Template)

Problem Statement (PS)

| Problem Statement (PS) | I am (Customer) | I want to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | A cricket coach | Monitor player performance across different matches | However, it is difficult to obtain accurate statistics quickly | The data is saved in multiple files and formats | Frustrated and uncertain while selecting the team |
| PS-2 | A team manager | Keep detailed records of every player | However, updating and validating the information takes a lot of time | Most of the data is recorded manually | Stressed and less productive |
| PS-3 | A sports analyst | Study player performance trends and prepare reports | However, comparing player statistics is challenging | There is no single centralized analytics system | Confused and dissatisfied |

## 2.2Empathy Map Canvas

**1. Think and Feel**

This section focuses on what the user thinks and feels internally.

It includes:

User’s worries

Aspirations and dreams

Motivations

Concerns

What really matters to them

**2. See**

This section describes what the user sees in their environment.

It includes:

What they see around them

Products or services available

What competitors offer

What their friends or colleagues are doing

**3. Say**

This section records what the user says in interviews or conversations.

It may include:

Statements about their experience

Opinions about products

Feedback they give

**4. Do**

This section focuses on the actions and behaviors of the user.

It includes:

What the user actually does

Their habits and actions

How they interact with products or services

**5. Hear**

This section describes what the user hears from others.

It includes influences such as:

Friends

Family

Teachers

**6. Pain and Gain** (Sometimes Added Sections)

Pain

Pain refers to the problems, frustrations, and obstacles users face.

## 2.3 Brainstorming

Brainstorming is a creative thinking technique used to generate many ideas or solutions for a problem in a short time. Here are the key points about brainstorming:

- A method used to generate new ideas and solutions.
- Usually done in groups, but can also be done individually.
- Encourages free thinking and creativity.
- All ideas are accepted during the session, even unusual ones.
- No criticism or judgment is allowed while ideas are being shared.
- Participants are encouraged to build on others' ideas.
- The main goal is to produce as many ideas as possible.

# 3.REQUIREMENT ANALYSIS PHASE

## 3.1 Customer Journey Map

A Customer Journey Map is a visual representation that shows the steps a customer takes when interacting with a product or service. It helps understand the customer's experience, needs, and problems at each stage.

In the requirement analysis phase, the customer journey map is used to identify user expectations, pain points, and opportunities for improvement. It helps developers and analysts design a system that meets customer needs and improves user experience.

## 3.2 Solution Requirements

### Functional Requirements

Functional requirements define the specific functions or operations that the system must perform. They describe what the system should do to meet the needs of users and achieve the system's objectives.

These requirements include activities such as user login, data processing, storing information, generating reports, and managing user actions. Functional requirements focus on the inputs, processes, and outputs of the system.

## Non-Functional Requirements (NFRs)

### NFR-1: Usability

Usability refers to how easy and convenient it is for users to use the CRICKSTATS system. It ensures that users can quickly understand the interface, navigate the system easily, and access cricket information without confusion.

### Example:

In CRICKSTATS, a user can quickly check live match scores, search for a player's statistics, or view team rankings with just a few clicks because the interface is simple and user-friendly.

## NFR-2: Security

### Definition:

Security refers to the protection of the system and its data from unauthorized access, misuse, or modification. It ensures that only authorized users can access or manage cricket data.

### Example:

In CRICKSTATS, only the administrator can log in and update player statistics or match scores, while normal users can only view the information. This protects sensitive player data from unauthorized access.

## NFR-4: Performance

### Definition:

Performance refers to the speed and efficiency of the system in processing data and responding to user requests. The system should provide information quickly without delays.

### Example:

In CRICKSTATS, when a user opens the live match score page, the system should load the scores and player statistics quickly, even when many users are accessing the system at the same time.

## NFR-5: Availability

### Definition:

Availability refers to the ability of the system to be accessible and operational for users whenever they need it.

### Example:

In CRICKSTATS, users should be able to check live match scores and player statistics at any time, especially during live cricket matches, without the system being down.

## NFR-6: Scalability

### Definition:

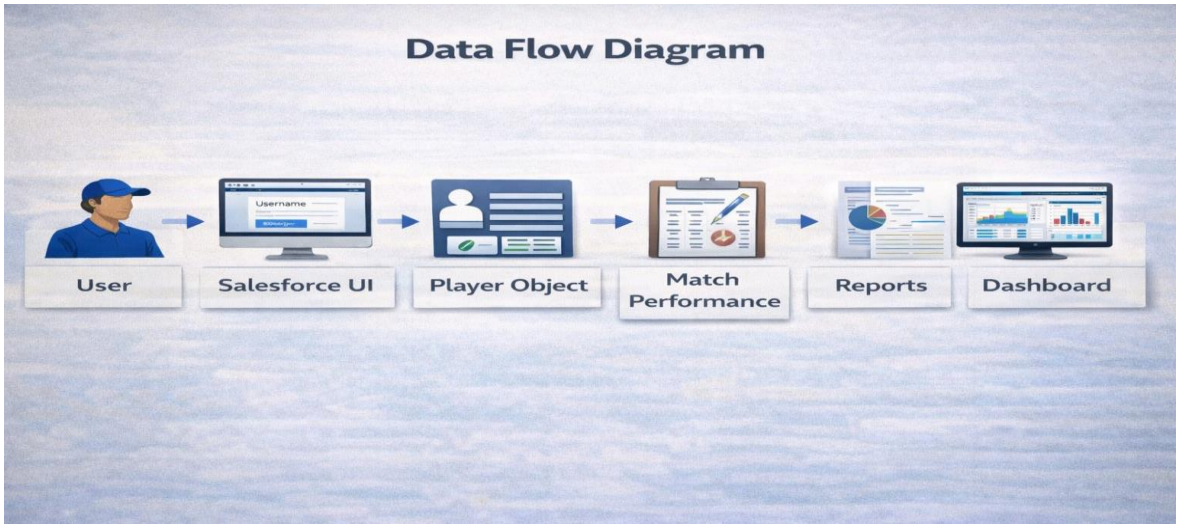
Scalability refers to the ability of the system to handle an increasing number of users, matches, and data without reducing performance.

### Example:

In CRICKSTATS, during a major tournament like the ICC Cricket World Cup, many users may access the system at the same time. The system should handle thousands of users checking live scores and statistics without slowing down

## 4.Data Flow Diagram

### Data Flow




User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard

# 5.Project Design Phase – Problem Solution Fit

Problem–Solution Fit in the project design phase means ensuring that the CRICKSTATS system is designed to solve the main problems faced by cricket fans and users when accessing match information.

**Problem:**

- Cricket fans find it difficult to get quick and accurate live scores and player statistics.
- Information may be scattered across different platforms.
- Users need a simple system to view match details, player performance, and team statistics.

**Solution:**

Develop a centralized system that provides live match scores, player statistics, and team details in one place.

Design a user-friendly interface so users can easily navigate the system.

Provide fast updates and accurate data for matches and players.

**Example:**

A user opens CRICKSTATS and can quickly check live match scores, view player batting and bowling statistics, and see team rankings in a single platform.

**Proposed Solution – CRICKSTATS**

The proposed solution is to develop the CRICKSTATS system, which provides a centralized platform for cricket information. This system helps users easily access live match scores, player statistics, team details, and match results in one place.

Key Points:

- Provide real-time live match score updates.
- Store and display player batting and bowling statistics.
- Show team information and rankings.
- Allow administrators to update match data and player records.
- Offer a simple and user-friendly interface for easy navigation.
- Ensure fast access to cricket information for fans and users.

**Example:**

A user can open CRICKSTATS to check live scores, view a player's performance, and see match results quickly and easily.

# 6. Project Development Phase:

## Project Flow

Milestone 1 : Creation of developer account

Milestone 2 : Object Creation

Milestone 3 : Tabs

Milestone 4 : The Lightning App

Milestone 5 : Fields & Relationships

Milestone 6 : Roll-up Summary

Milestone 7 : Flows

# Implementation

## Milestone 1 - Salesforce developer account creation :

## Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup

2. On the sign up form, enter the following details :

1) First name & Last name

2) Email

3) Role : Developer

4) Company : College or Company Name

5) County : India

6) Postal Code : pin code

7) Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in that format : username@organization.com

Click on sign me up after filling these.

**Activity 2: Account Activation :**

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.

2. Click on Verify Account.

3. Give a password and answer a security question and click on change password.

4. Then you will redirect to your salesforce setup page.

**Milestone 2 - Object Creation**

Activity 1:Player Object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.

2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.

3. On the Object Manager page, click Create | Custom Object.

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- **Label** : Player
- **Plural Label** : Players
- Enter Record Name Label and Format

  Record Name : Player Name

  Data Type **:** Text
- **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status :** check the box for Allow Search.
- Save

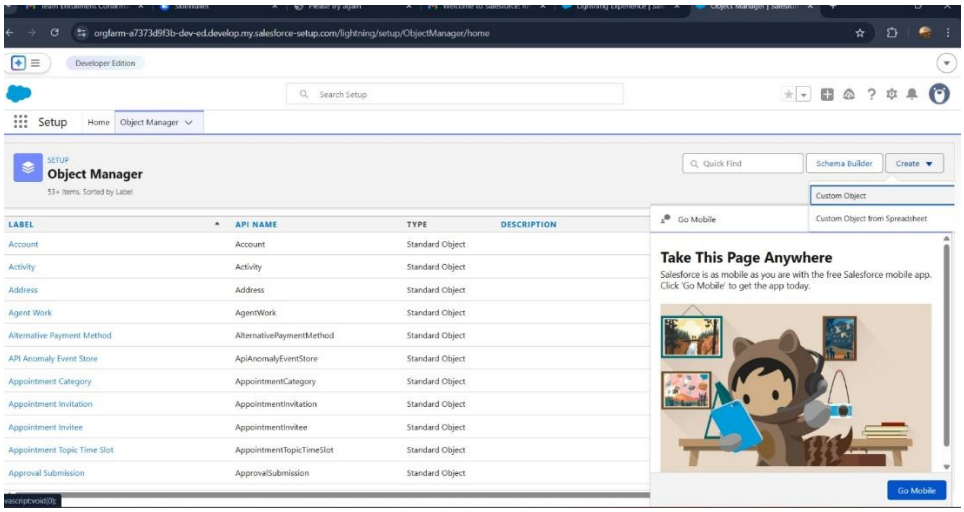
Activity 2 : Match Performance Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- **Label** : Match Performance
- **Plural Label** : Match Performances
- Enter Record Name Label and Format

  Record Name : Customer Account Name

  Data Type **:** Text
- **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
- **Search Status :** check the box for Allow Search. ● Save.

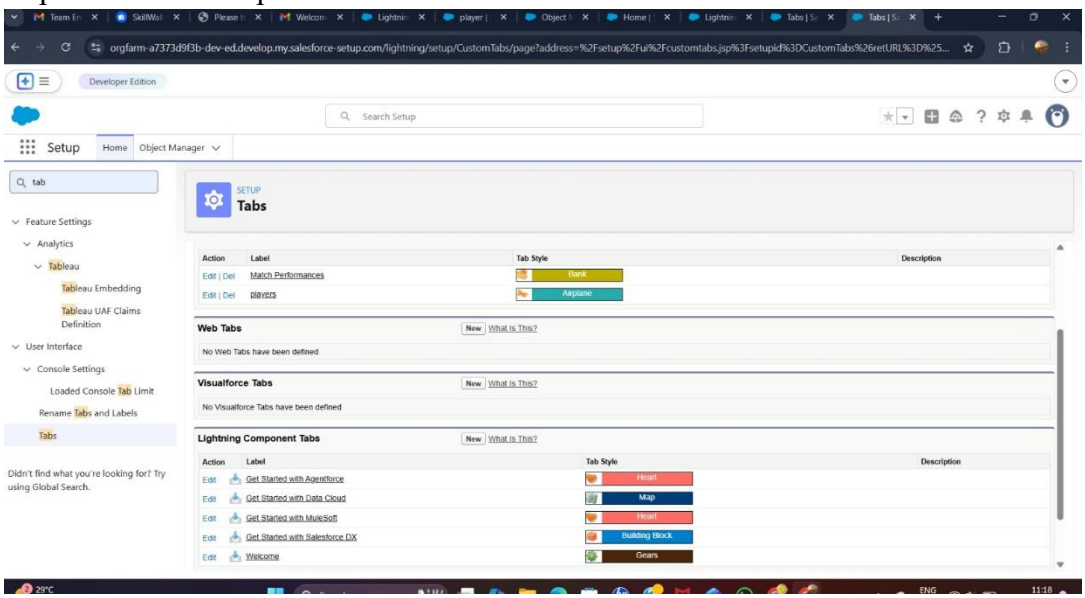



**Milestone 3 - Tabs**

Activity 1 : Custom tabs creation

Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)

a. Select Object(Player) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab .
b. Make sure that the Append tab to users' existing personal customizations is checked.
c. Click save.
d. Repeat the same steps for Match Performances.




**Milestone 4 - The Lightning App**

Activity 1 : Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.

2. Fill the app name in app details and branding as follow
   App Name : CRICKSTATS
   Developer Name : Auto Populates
   Add image optional (if you want to give any image you can otherwise not mandatory)
   - Add Primary color Hex or leave it to default. Then click Next.

3. In App options
   Navigation style : Standard navigation
   Setup experience : setup
   Supporters form factors : Desktop and phone
   Then click Next

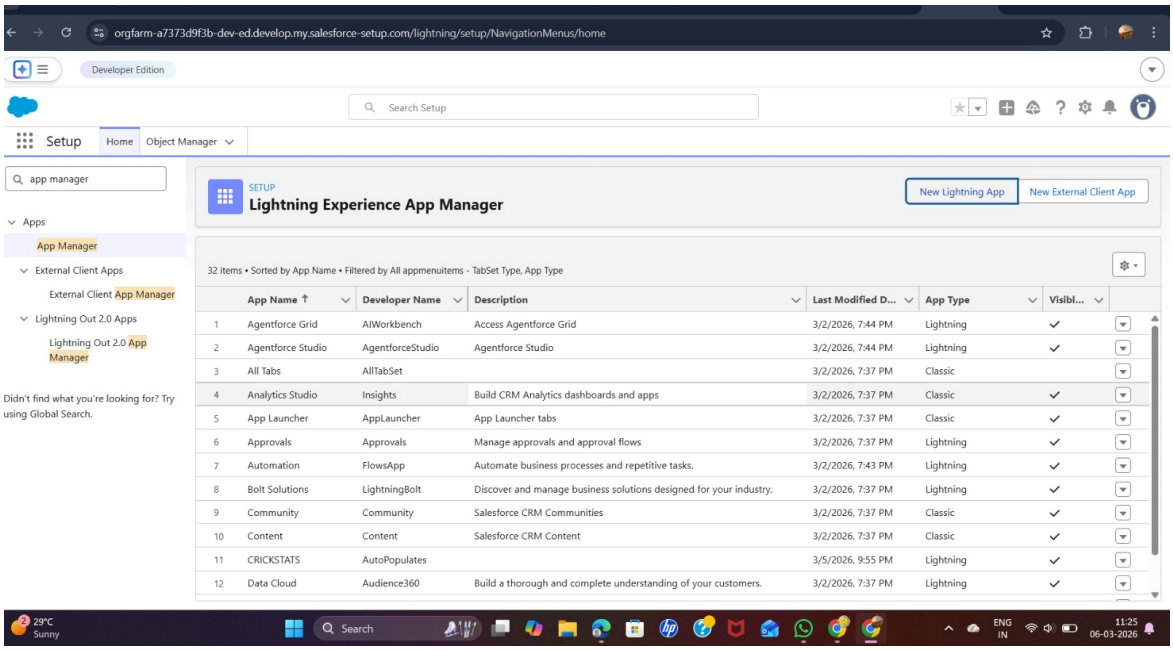
4. In Utility items

Utility Bar alignment : Default Then click
Next.

5. Navigation items
   Select the created Custom Objects and required standard objects
   - Players
     Match Performances
     Reports
     Dashboards

6. To Add User Profiles : System Administrator

7. Click Save & Finish.

8. 


**Milestone 5 : Fields & Relationships**

Activity 1 : Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.
6. Next, choose a data type as Text. Choosing a data type helps you format the field input.

7. Click on next.

8. Data Type : **Text**
   Field Label: Player Name
   Field Name : Auto Populates
   Length : 50
   Click on Next → Next → Save and new.

9. Repeat the Same steps for remaining fields

   - Data Type : **Text**
     Field Label: Player Number
     Field Name : Auto Populates
     Length : 3
     Click on Next → Next → Save and new.

   - Data Type : **Rich Text Area**
     Field Label : Profile Picture
     Field Name : Auto Populates
     Click on Next → Next → Save and new.

   - Data Type : **Date**
     Field Label : Date of Birth
     Field Name : Auto Populates
     Click on Next → Next → Save and new.

   - Data Type : **Picklist**
     Field Label : Role
     Field Name : Auto Populates
     Value : Enter value, with each value separated by a new line. Batsman
     Bowler
     Batsman & Wicket Keeper
     All Rounder

   - Data Type : **Picklist**
     Field Label : Batting Style
     Field Name : Auto Populates
     Value : Enter value, with each value separated by a new line.
     Right- Handed
     Left - Handed
     Click on Next → Next → Save and new.

   - Data Type : **Picklist**
     Field Label : Bowling Style
     Field Name : Auto Populates
     Value : Enter value, with each value separated by a new line.
     Right-Arm Fast

Left-Arm Fast
Right-Arm Medium
Left-Arm Medium
Right-Arm Spin
Left-Arm Spin

Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Nationality
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line.
  Australia
  Bangladesh England
  India
  New Zealand
  Pakistan
  South Africa Sri
  Lanka
  West Indies
  Afghanistan
  Zimbabwe
  Ireland
  Scotland
  Netherlands
  United Arab Emirates
  Nepal
  United States of America

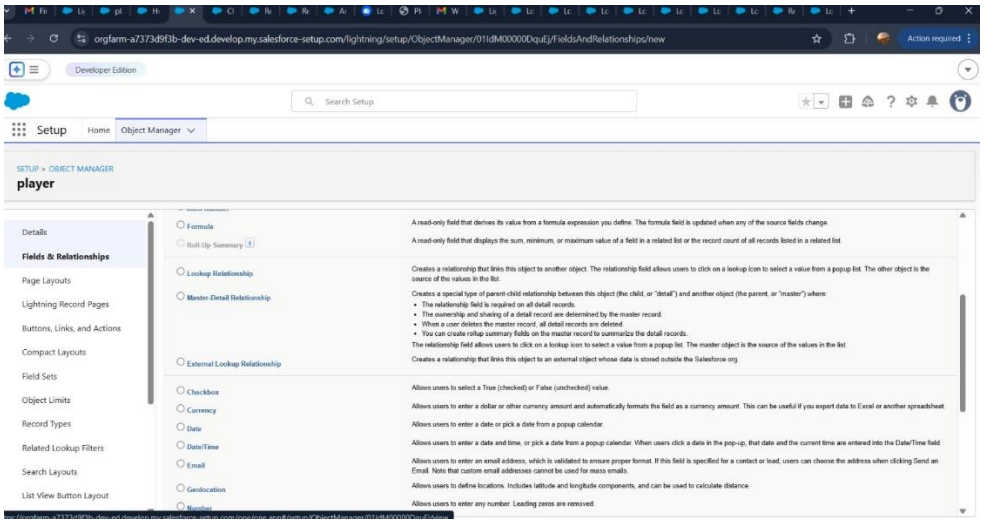
  Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Team
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line
  Chennai Super Kings (CSK)
  Delhi Capitals (DC)
  Kolkata Knight Riders (KKR)
  Mumbai Indians (MI)
  Punjab Kings (PBKS)
  Rajasthan Royals (RR)
  Royal Challengers Bangalore (RCB)
  Sunrisers Hyderabad (SRH)
  Gujarat Titans (GT)
  Lucknow SuperGiants (LSG)
- Data Type : **Formula**
  Field Label : AVG Score
  Field Name : Auto Populates
  Formula Return : **Number**
  Formula : IF(

Matches_Played__c > 0,
Total_Runs__c / Matches_Played__c ,
0
)

Click on Next → Next → Save & New.

- Data Type : **Lookup Relationship**
  Related To : Match Performance
  Field Label : Match Performance
  Field Name : Auto Populates
  Click on Next → Next → Save.




## Activity 2 : Create Custom fields for Match Performance object.

1. Data Type : **Text**
   Field Label: Match Performance Name
   Field Name : Auto Populates
   Length : 80
   Click on Next → Next → Save and new.

2. Data Type : **Checkbox** Field Label: Player of the Match Field Name : Auto Populates
   Click on Next → Next → Save and new.

3. Data Type : **Picklist**
   Field Label: Match Result
   Field Name : Auto Populates
   Value : Enter value, with each value separated by a new line
   WON
   LOST
   DRAW
   Click on Next → Next → Save and new.

4. Data Type : **Number**

Field Label: Runs Scored
Field Name : Auto Populates
Click on Next → Next → Save and new.

5. Data Type : **Number**
Field Label: Balls Faced
Field Name : Auto Populates
Click on Next → Next → Save and new.

6. Data Type : **Number**
Field Label: Fours
Field Name : Auto Populates
Click on Next → Next → Save and new.

7. Data Type : **Number**
Field Label: Sixes
Field Name : Auto Populates
Click on Next → Next → Save and new.

8. Data Type : **Number**
Field Label: Overs Bowled
Field Name : Auto Populates
Click on Next → Next → Save and new.

9. Data Type : **Number**
Field Label: Wickets Taken
Field Name : Auto Populates
Click on Next → Next → Save and new.

10. Data Type : **Number**
Field Label: Runs Conceded
Field Name : Auto Populates
Click on Next → Next → Save and new.

11. Data Type : **Number**
Field Label: Maiden Overs
Field Name : Auto Populates
Click on Next → Next → Save and new.

12. Data Type : **Number**
Field Label: Stumpings
Field Name : Auto Populates
Click on Next → Next → Save and new

13. Data Type : **Number**
Field Label: Catches
Field Name : Auto Populates
Click on Next → Next → Save and new.

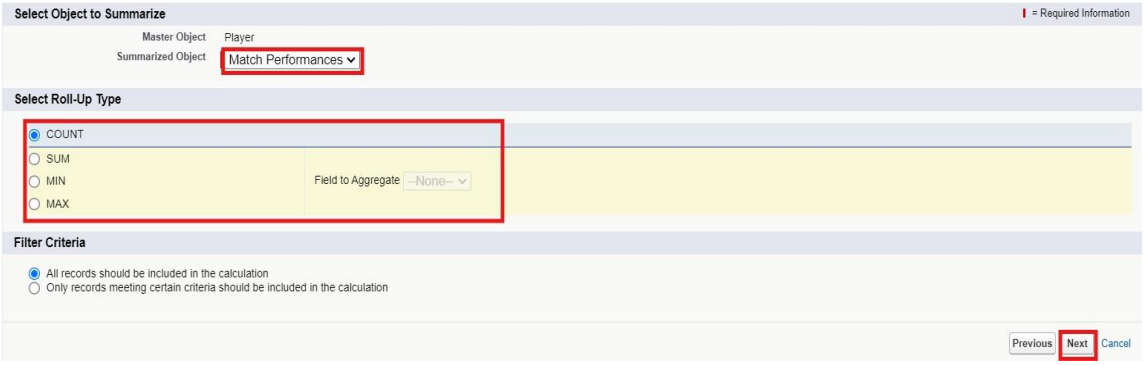
14. Data Type : **Master - Detail Relationship**
    Related To : Player
    Field Label : Player
    Field Name : Auto Populates Click on Next
    → Next → Save.

**Milestone 6 : Roll-up Summary**

## Activity 1 : Create Roll-up Summary for Player object.

Note :- Roll-up Summary can only be created on parent object in Master- Detail Relationship.

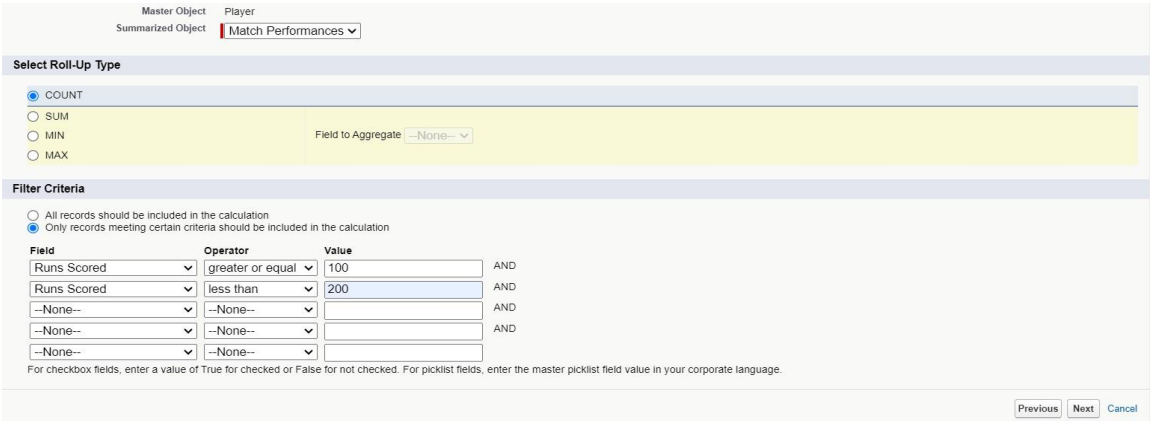
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist. Click Next,
6. Choose a data type as **Roll-up Summary**. Click Next.
   Field Label: Matches Played
   Field Name : Auto Populates
   Summarized Object : Match Performances
   Select Roll-up Type : COUNT




Filter Criteria : All records should be included in the calculation. Click on Next
→ Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Number of 50's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : COUNT
  Filter Criteria : Only records meeting certain criteria should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Number of 100's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : COUNT
  Filter Criteria : Only records meeting certain criteria should be included in the calculation.
  Click on Next → Next → Save and new.




- Data Type : Roll-up Summary Field Label:
  Total 4's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Fours
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Total 6's
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Sixes
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Total Runs
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Runs Scored
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Total Overs
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Overs Bowled
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Total Wickets
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Wickets Taken
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

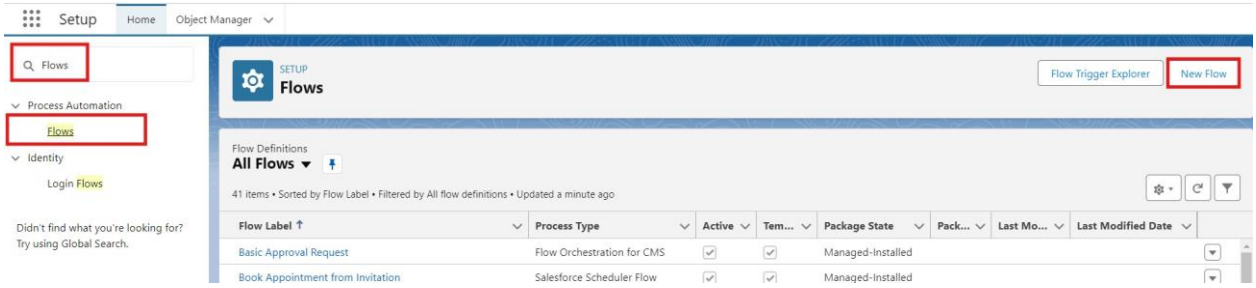
- Data Type : Roll-up Summary Field Label:
  Total Stumpings
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Stumpings
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

- Data Type : Roll-up Summary Field Label:
  Total Catches
  Field Name : Auto Populates
  Summarized Object : Match Performances
  Select Roll-up Type : SUM
  Field To Aggregate : Catches
  Filter Criteria : All records should be included in the calculation.
  Click on Next → Next → Save and new.

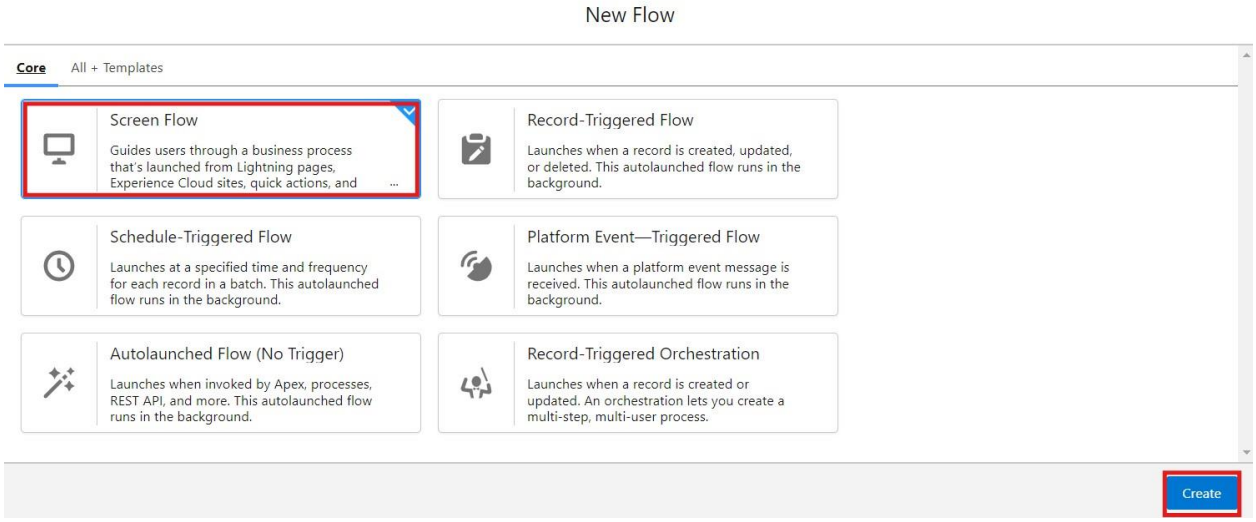
## Milestone 7 : Flows

### Activity 1 : Screen Flow

1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.




2. Select Screen Flow and click Create.

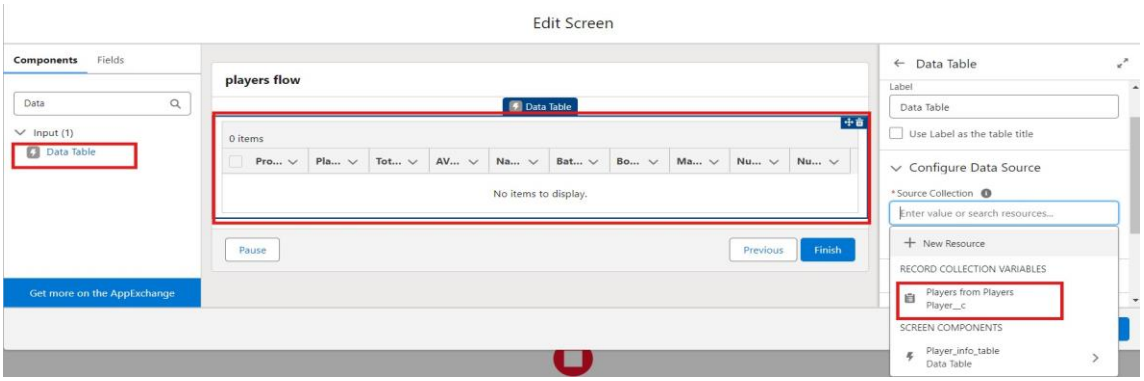



3. Click (+) Icon on the flow builder and Select **Screen Component.**
   **Label :** Players screen
   **API** : Players_screen
   Select the **Picklist** component and place it on the screen.
   **Label :** Team
   **API :** Team
   From Component Pane on the left side select **Component Type** : Picklist
   **Data Type** : Text
   **Choice** : Click New Choice Resources
   **Resource Type** : Picklist Choice Set
   **API Name :** Team
   **Object :** Player
   **Data Type** : Picklist **Field** :
   Team Click Done.

**4.** Click (+) Icon on the flow builder and Select **Get Records Component.**
   **Label :** Players
   **API Name :** Players
   **Object** : Player
   **Filter player records** :
   **Field** : Team
   **Operator** : Equals
   **Value** : Click New Resource

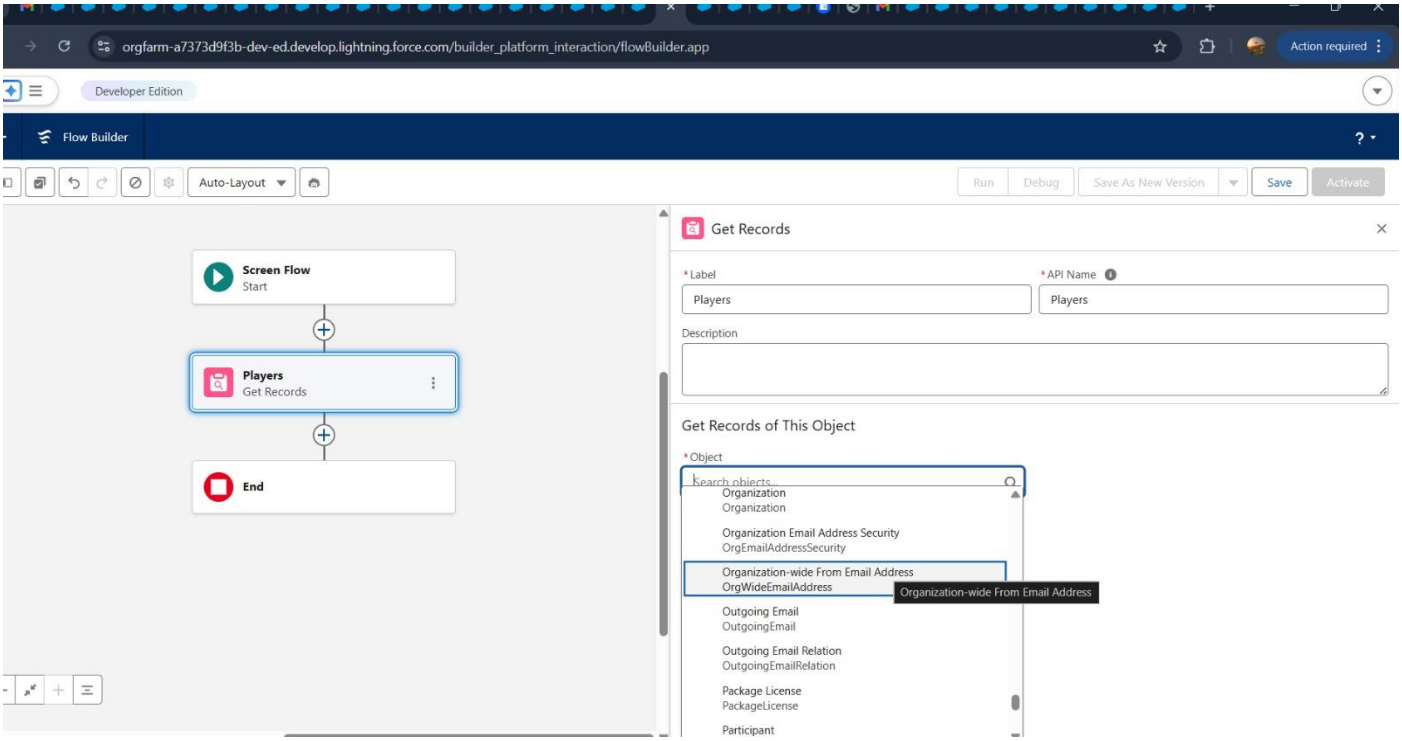
**Resource Type** : Variable
**API Name** : Dynamic
**Data Type** : Text
**Default Value : Screen component :** Team (Picklist)

5. Click (+) Icon on the flow builder and Select **Screen Component.**
**Label :** Player_info
**API Name :** Player_info
From Component Pane on the left side select **Component Type :** Data table
**API Name** : Player_info_table
**Label** : Data Table
**Configure Data Source : Source Collection** : Players from players (Player__c)



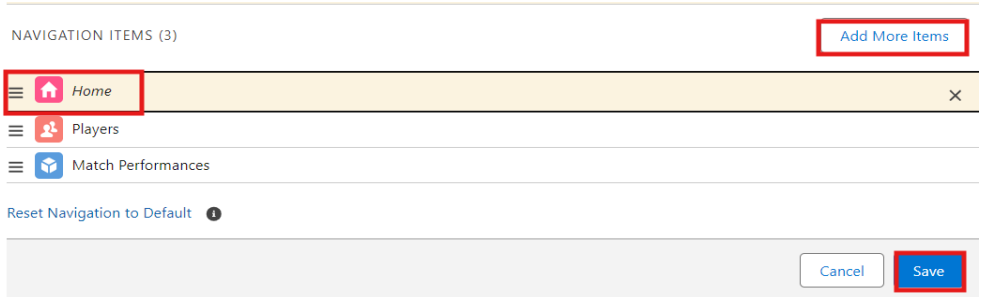

**In Configure Columns : Add all the required Columns**
Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling Style | Matches Played | Number of 50's | Number of 100's Click Done.

Click Save.
**Flow Label** : Player Flow
**Flow API Name** : Player_Flow

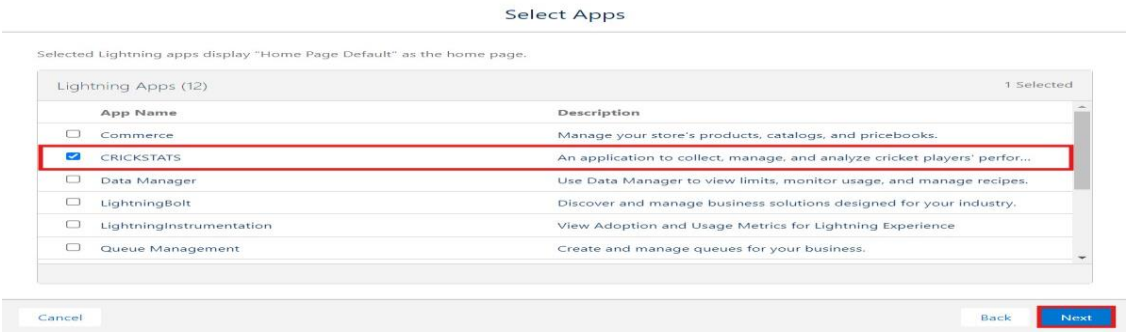



Activity 2 : Flow on Home Page

1. On your CRICKSTATS Application on the top right of the Navigation Bar click on **Personalize your nav bar icon (Pencil Icon)** and Add Home Page and click Save.




2. Remove all other components and Add **FLOW** component from the standard components on the left side.
3. **Flow** : Players Flow
4. Click **Save** and **Activation**.
5. Click on **App Default** and select **Assign to Apps.**
6. Select **CRICKSTAS** App and Click **Next** and then click **Save**.



## 7.FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency. The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics.

In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of generated reports and interactive dashboards were documented to confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deploys

## 8. ADVANTAGES & DISADVANTAGES

### Advantages :

- **Easy Access to Player Statistics**

Crickstats allows users to easily view player details like runs, wickets, strike rate, and averages.

- **Accurate Data Management**

The system stores cricket match data accurately and reduces manual calculation errors.

- **Quick Match Analysis**

Users can quickly analyze match results, team performance, and player statistics.

- **Time Saving**

Automatic calculation of scores and statistics saves time compared to manual record keeping.

### Disadvantages

- **Dependence on Technology**

The system works only with computers or internet access. Without technology, it cannot be used.

- **Data Entry Errors**

If wrong data is entered (runs, wickets, overs), the statistics and reports will also be incorrect.

- **Maintenance Cost**

The system may require regular updates, maintenance, and technical support.

- **Security Risks**

Cricket data may be lost or hacked if proper security measures are not used.

## 9.COCLUSION:

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

## 10.FUTURE SCOPE

- **Integration with Live Match Updates**
  In the future, Crickstats can provide **real-time match scores and live statistics** during cricket matches.

- **Mobile Application Development**
  A mobile app can be developed so users can **check player stats and match details anytime and anywhere**.

- **Advanced Data Analytics**
  The system can include advanced analytics to **analyze player performance, team strategies, and match predictions**.

## 11. APPENDIX

- Salesforce Objects & Fields
- Manual match statistics
- Flow Configurations